Do LLMs show systematic instability under paraphrasing, and is it caused by paraphrase sensitivity or boundary ambiguity?

ST5230 Applied Natural Language Processing

Group21 Lin Ting

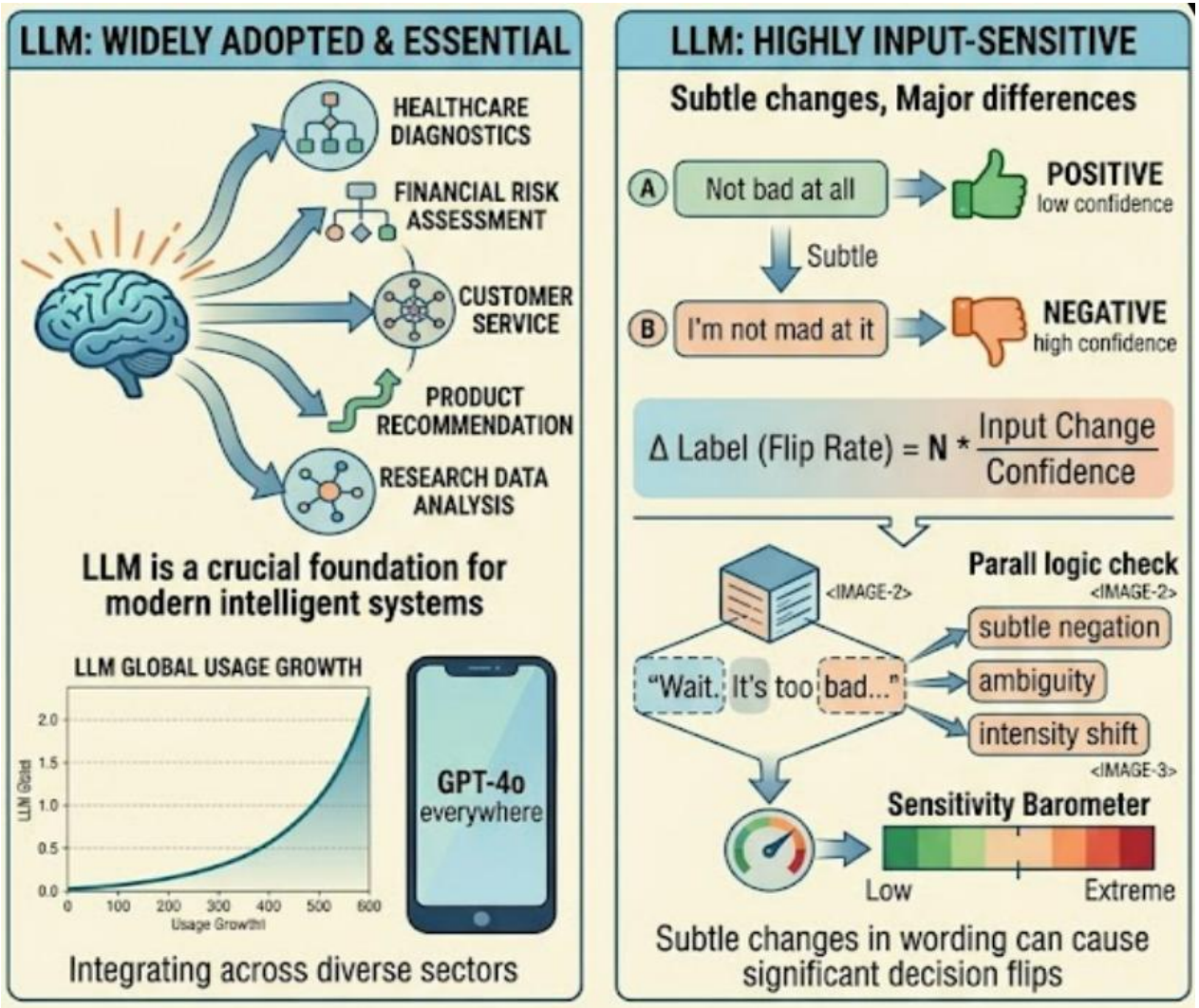
NUS
National University of Singapore

Motivation

- **LLM widely used**
- **Input-sensitive**

Question:

Do semantically equivalent paraphrases lead to different predictions?



Research Questions

Does paraphrasing cause label flips?

Are flips random or structured?

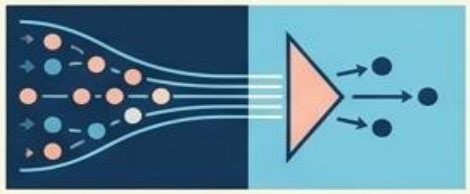
Do models behave differently?

Pipeline Overview

Text → Paraphrase → Model Classification → Compare & Analysis → Conclusion

LLM Sentiment Classification Robustness Analysis Pipeline

Phase 1: DATA PREPARATION



Sample 200-300 Yelp Reviews (*Source*: `yelp_review_full`)
Cleaning: Remove Long Text & Multi-Sentence Reviews
Result: `yelp_300.csv`

Phase 2: PARAPHRASE GENERATION

LLM: GPT-4o
Task: Generate 1-2 Paraphrases per Review
Result:
`yelp_with_paraphrase.csv`

Phase 3: PARAPHRASE FILTERING

Similarity
Layer 1: Semantic Embedding (Fast)
LLM: all-MiniLM-L6-v2
Method: Cosine Similarity
Threshold: > 0.85

Evaluator
Layer 2: LLM Validation (Accurate)
LLM: GPT-4o-mini
Strict Criterion: YES/NO for same meaning and sentiment (intensity preserved)

Result: `final_validated_dataset.csv`
(200 High-Quality Pairs)

Phase 4: SENTIMENT CLASSIFICATION EXPERIMENT

Cross-Model Evaluation
(GPT-4o-mini vs Mistral-7B)
determinslic reasoning: 'temperature=0'

Input:
{text} & {paraphrase}

Sampling:
n=5 times per sample to calculate confidence

Experiment Record Table

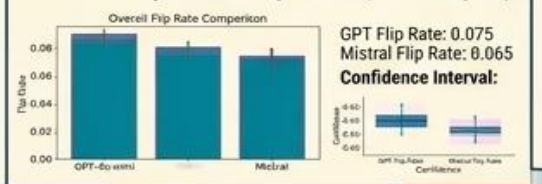
gpt_orig_labels	gpt_para_labels	gpt_flip
gpt_orig_labels	gpt_para_labels	mis_orig_labels
gpt_flip	text	mis_orig_labels

Fragment Distribution

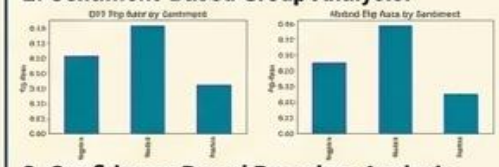
gpt_orig_label	error_type	error_type
I used to go often ...	shift	Negation change
I will not return ...	shift	Intensity shift
I will not return ...	shift	Intensity shift
I used to go often ...	shift	Intensity shift
I used to return ...	shift	Intensity shift

Phase 5: MULTI-LEVEL ANALYSIS

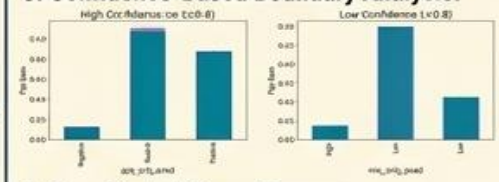
1. Overall Flip Rate Comparison (Bootstrap CI):



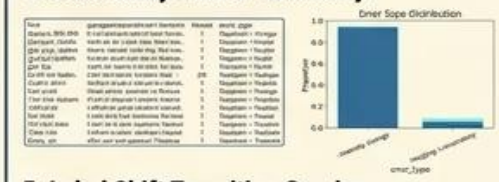
2. Sentiment-Based Group Analysis:



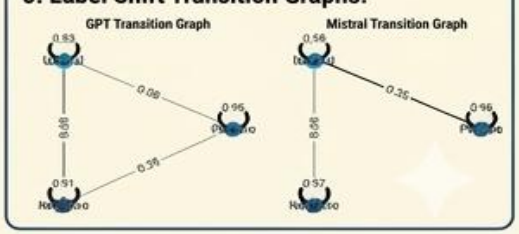
3. Confidence-Based Boundary Analysis:



4. Case Study & Error Taxonomy:



5. Label Shift Transition Graphs:



Phase 6: REPORTING & INSIGHTS

- Discuss Architecture/Scale Effects on Robustness
- Propose Reliability Guidelines for LLM Deployment in Decision Systems

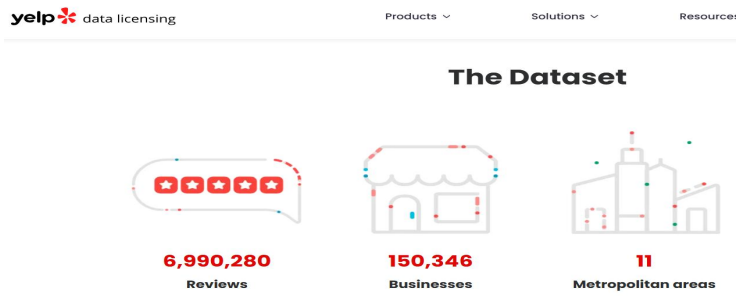
Data Preparation

Source: Yelp Review Dataset

1	text	label	paraphrase
2	ery importan	4	have emphasized as
3	vants to deal	1	the time or the pat
4	t I found out	3	vered I have fibroic
5	l be missed v	3	He will be greatly
6	ent. 90% of t	0	nen you call, there'
7	we have bee	4	ldberg is highly acc

original dataset

generated paraphrases



code for paraphrase generating using gpt-4o model:

```
from openai import OpenAI
import pandas as pd

client = OpenAI(api_key="sk-proj-[redacted]

df = pd.read_csv("Source_data_yelp_300.csv")

def paraphrase(text):
        response = client.chat.completions.create(
                model="gpt-4o",
                temperature=0.7,
                messages=[
                        {"role": "user", "content": f"""
Rewrite the following sentence by changing the wording while strict
Do not make it more positive or more negative. Keep the same int

Sentence:
{text}

Paraphrase:
"""}
                ]
        )
        return response.choices[0].message.content.strip()

df["paraphrase"] = df["text"].apply(paraphrase)

df.to_csv("yelp_with_paraphrase.csv", index=False)
```

Paraphrase generating prompt:

Rewrite the following sentence by changing the wording while strictly preserving its original meaning and sentiment.

Do not make it more positive or more negative. Keep the same intensity.

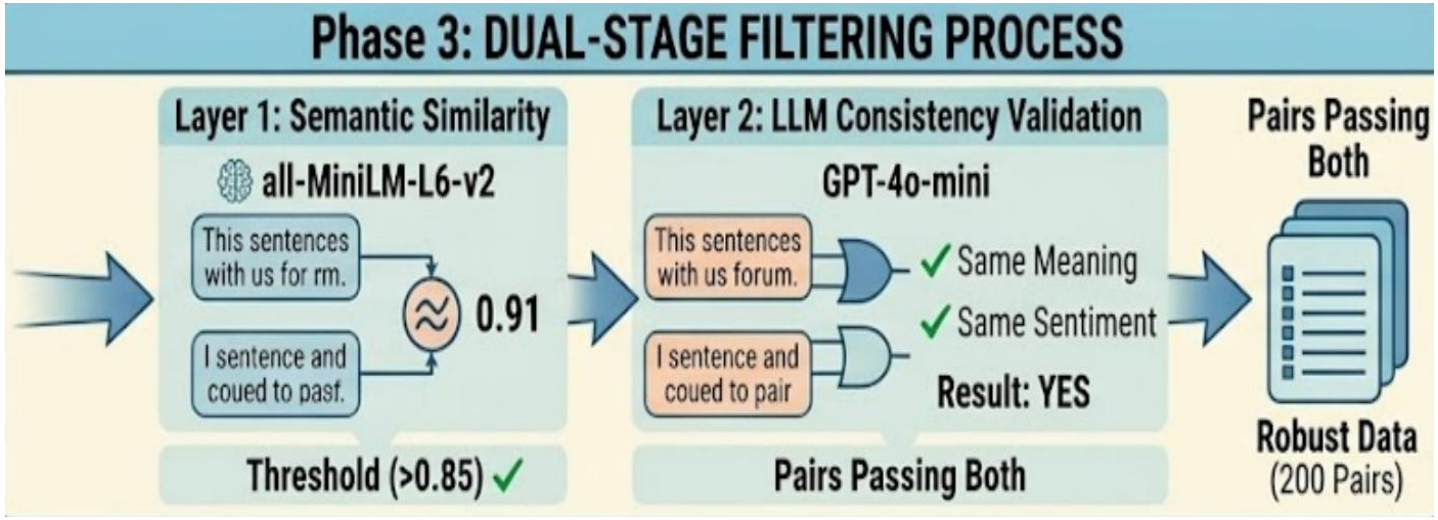
Data Preparation

Filtering out the robust data through 2 steps:

1	text	label	paraphrase	similarity	llm_valid
2	ery importan	4	have emphasized as	0.8515856	YES
3	vants to deal	1	the time or the pat	0.94968337	YES
4	t I found out	3	vered I have fibroic	0.919402	YES
5	l be missed v	3	He will be greatly	0.95387936	YES
6	ent. 90% of t	0	hen you call, there'	0.9298493	YES
7	we have bee	4	ldberg is highly acc	0.9710021	YES

1.Embedding similarity

2.LLM Validation



Model Setup

For each sample:

Inputs: **Both the original text and its paraphrase.**

Models: **Use two models (GPT-4o-mini and Mistral) to perform sentiment classification on both inputs using a standardized prompt.**

Sampling: **Perform 5 iterations of sampling for each task.**

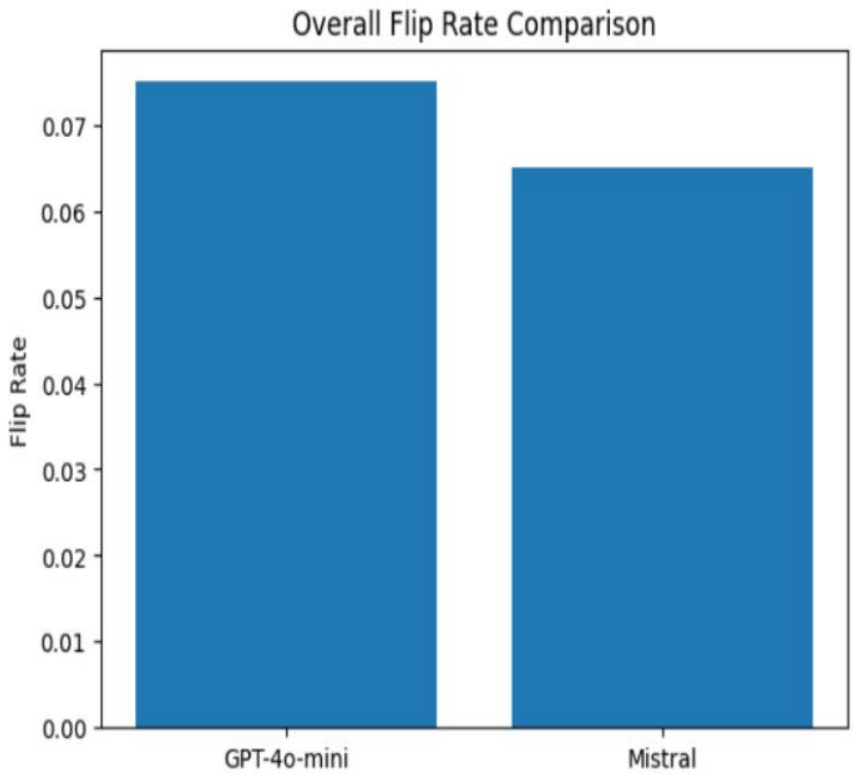
Output: **Calculate the flip rate and the confidence Interval.**

gpt_orig_labels	gpt_para_labels	mis_orig_labels	mis_para_labels	gpt_orig_conf	gpt_para_conf	mis_orig_c	mis_para_c	gpt_orig_pred	gpt_para_prs	mis_orig_prs	mis_para_pr	gpt_flip	mis_flip
e', 'Positive', 'Pve', 'Positive', 'Potive', 'Positive', 'Positive', 'Positive', 'Posi				1	1	1	1	Positive	Positive	Positive	Positive	FALSE	FALSE
e', 'Negative', 'Nve', 'Negative', 'Netive', 'Negative', 'Negative', 'Negative', 'Nega				1	1	1	1	Negative	Negative	Negative	Negative	FALSE	FALSE
e', 'Positive', 'Pve', 'Positive', 'Potive', 'Positive', 'Positive', 'Positive', 'Posi				1	1	1	1	Positive	Positive	Positive	Positive	FALSE	FALSE
e', 'Negative', 'Nve', 'Negative', 'Neutral', 'Negative', 'Negative', 'Negative', 'Nega				1	1	0.8	1	Negative	Negative	Negative	Negative	FALSE	FALSE
e', 'Negative', 'Nve', 'Negative', 'Netive', 'Negative', 'Negative', 'Negative', 'Nega				1	1	1	1	Negative	Negative	Negative	Negative	FALSE	FALSE
e', 'Positive', 'Pve', 'Positive', 'Potive', 'Positive', 'Positive', 'Positive', 'Posi				1	1	1	1	Positive	Positive	Positive	Positive	FALSE	FALSE

Overall Results

GPT Flip Rate: 0.075
Mistral Flip Rate: 0.065

GPT CI: [0.04 0.115]
Mistral CI: [0.035 0.105]



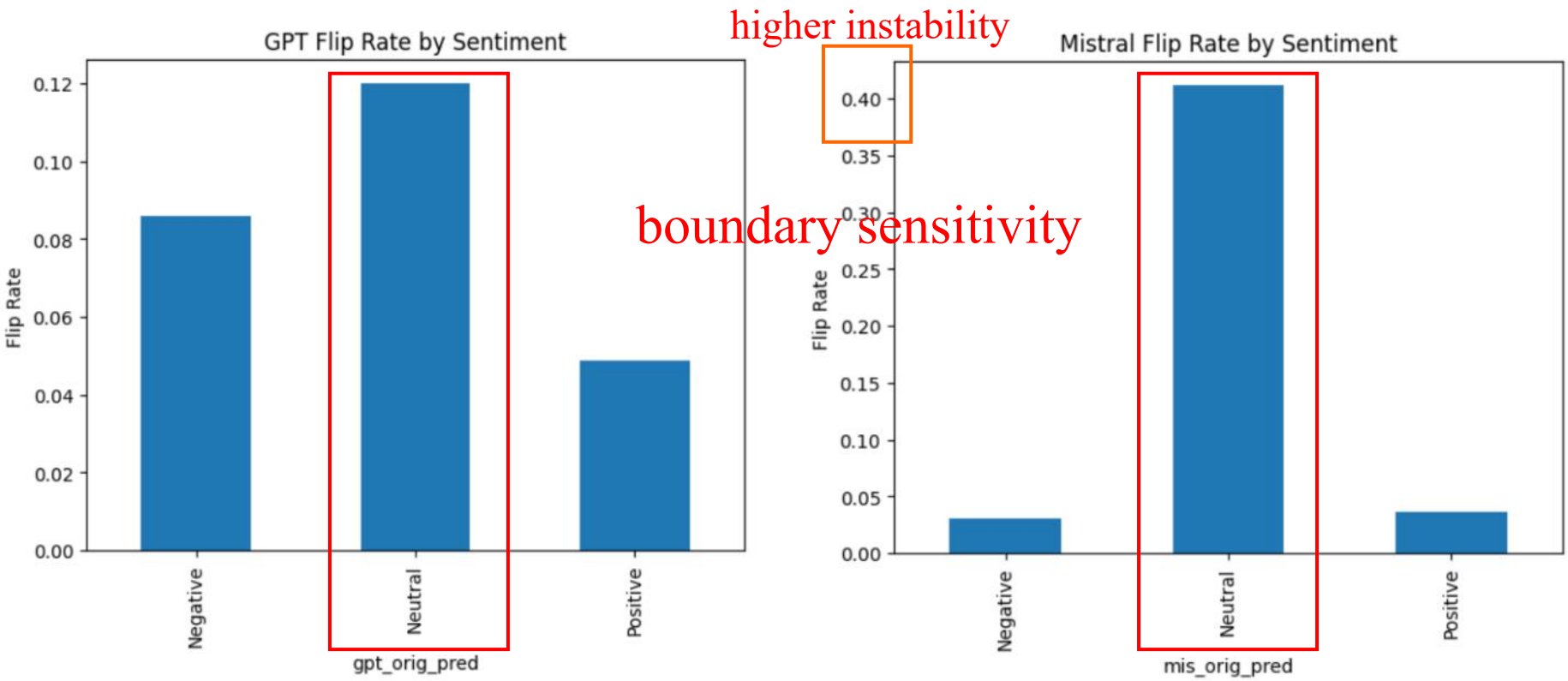
Flip Rate's Bootstrap CI

```python
import numpy as np

def bootstrap_ci(data, n=1000):
        samples = []
        for _ in range(n):
                sample = data.sample(frac=1, replace=True)
                samples.append(sample.mean())
        return np.percentile(samples, [2.5, 97.5])

gpt_ci = bootstrap_ci(df["gpt_flip"])
mis_ci = bootstrap_ci(df["mis_flip"])

print("GPT CI:", gpt_ci)
print("Mistral CI:", mis_ci)
```

```
GPT CI: [0.04 0.115]
Mistral CI: [0.035 0.105]
```

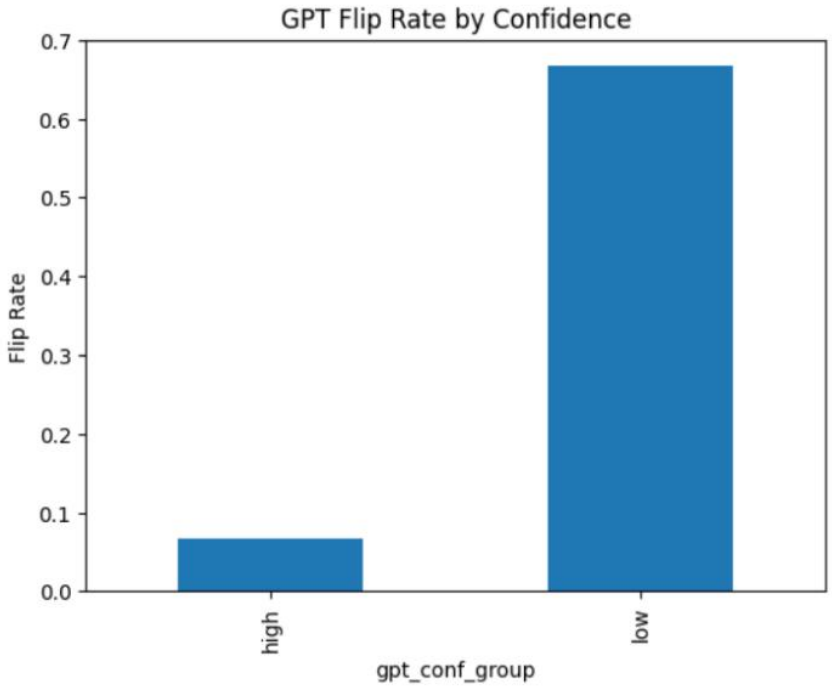
The flip rates of GPT-4o-mini and Mistral are similar, and the overlapping confidence intervals indicate that there is no statistically significant difference in their sensitivity to paraphrasing.

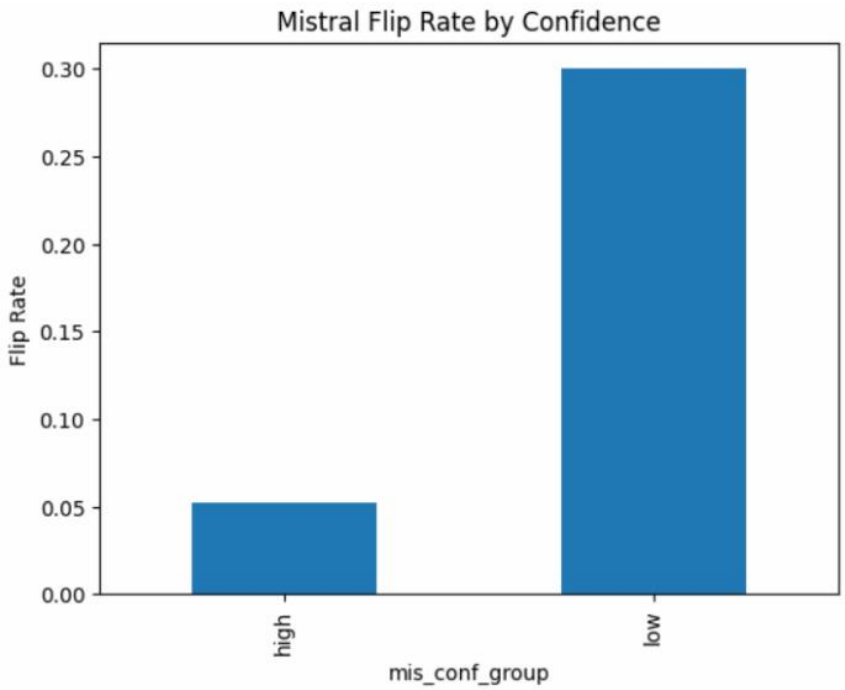
Flip Rate by Sentiment Results



Both models exhibit the highest Decision Flip Rate in the "Neutral" category, suggesting that decision instability is heavily concentrated near implicit sentiment boundaries. Notably, Mistral shows significantly higher sensitivity to paraphrasing in neutral cases (over 40%) compared to GPT-4o-mini.

Flip Rate by Confidence Results

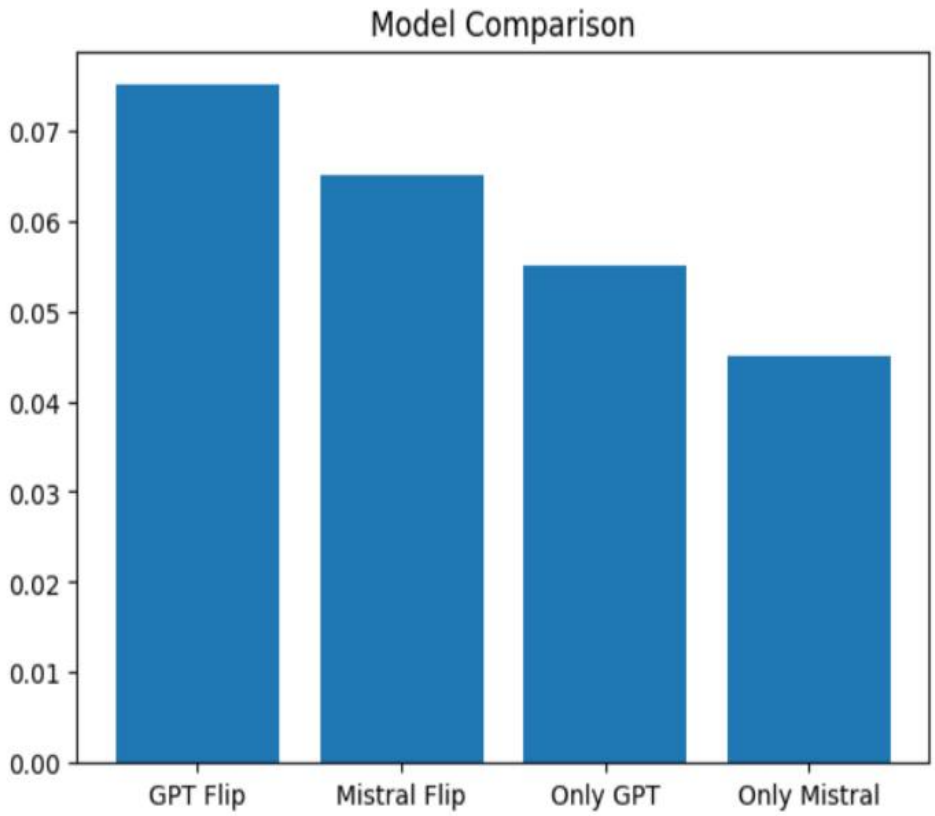




threshold = 0.8

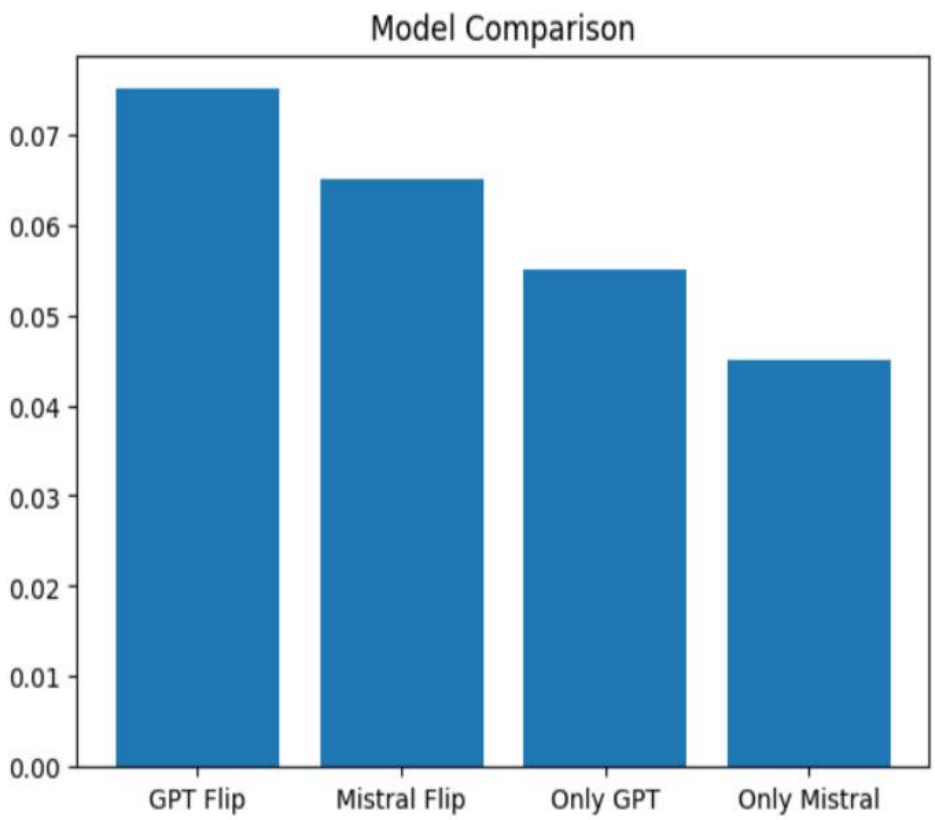
The relationship between confidence and stability:
Low-confidence predictions are much more likely to flip.
This suggests that many flips occur near decision boundaries.

Model Comparison



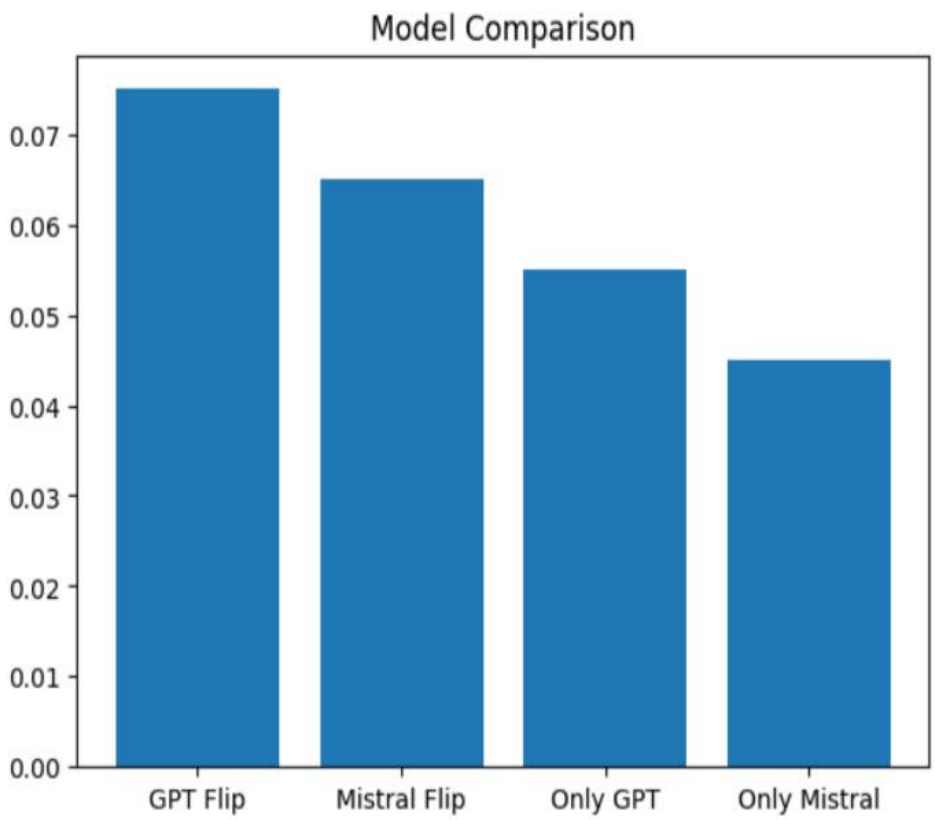
1. GPT-4o-mini
Demonstrates a slightly higher overall Decision Flip Rate compared to **Mistral**, indicating that even state-of-the-art proprietary models remain sensitive to minor phrasing variations.

Model Comparison



2. The presence of **"Only GPT" and "Only Mistral" flips** suggests that decision instability is not purely data-dependent but is also influenced by **specific model architectures and training biases.**

Model Comparison



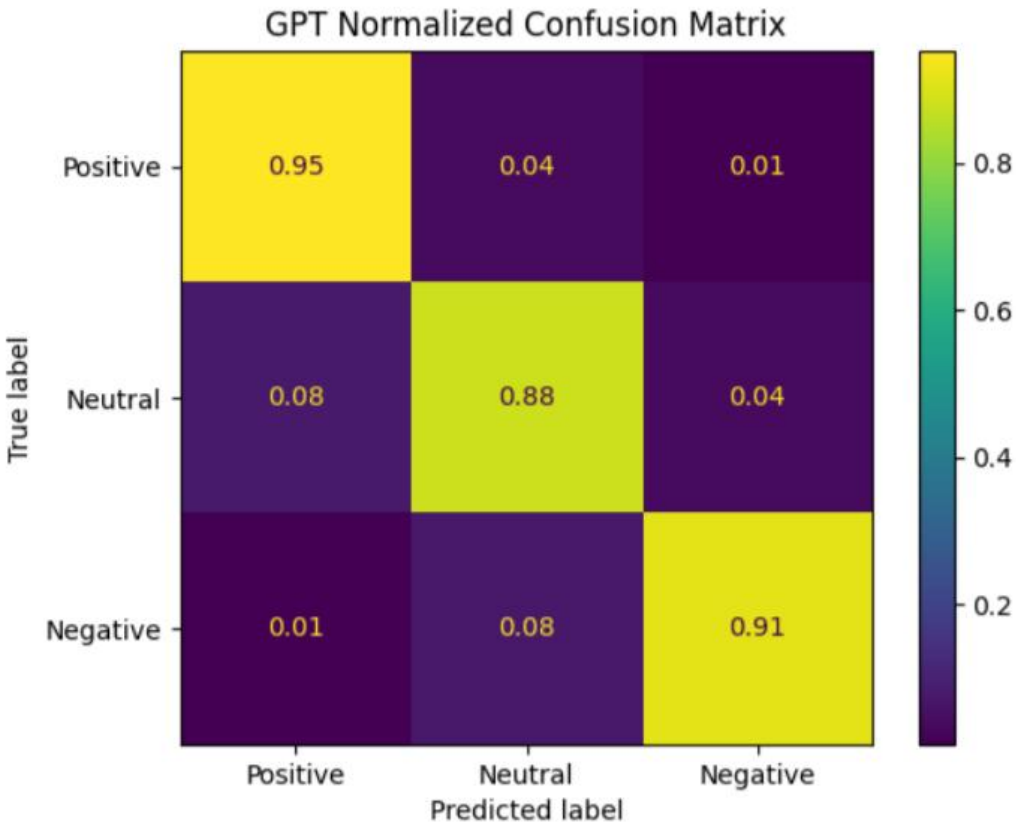
3. A significant portion of flips is **model-specific**, confirming that while some cases are inherently near decision boundaries, others trigger instability due to idiosyncratic linguistic sensitivities.

Confusion Matrix & Transition Graph

- The confusion matrix shows how predictions change.
- Most predictions remain consistent, but some shift across categories.

- The transition graph reveals that these changes are not random.
- Most transitions occur between adjacent classes, such as Positive to Neutral.

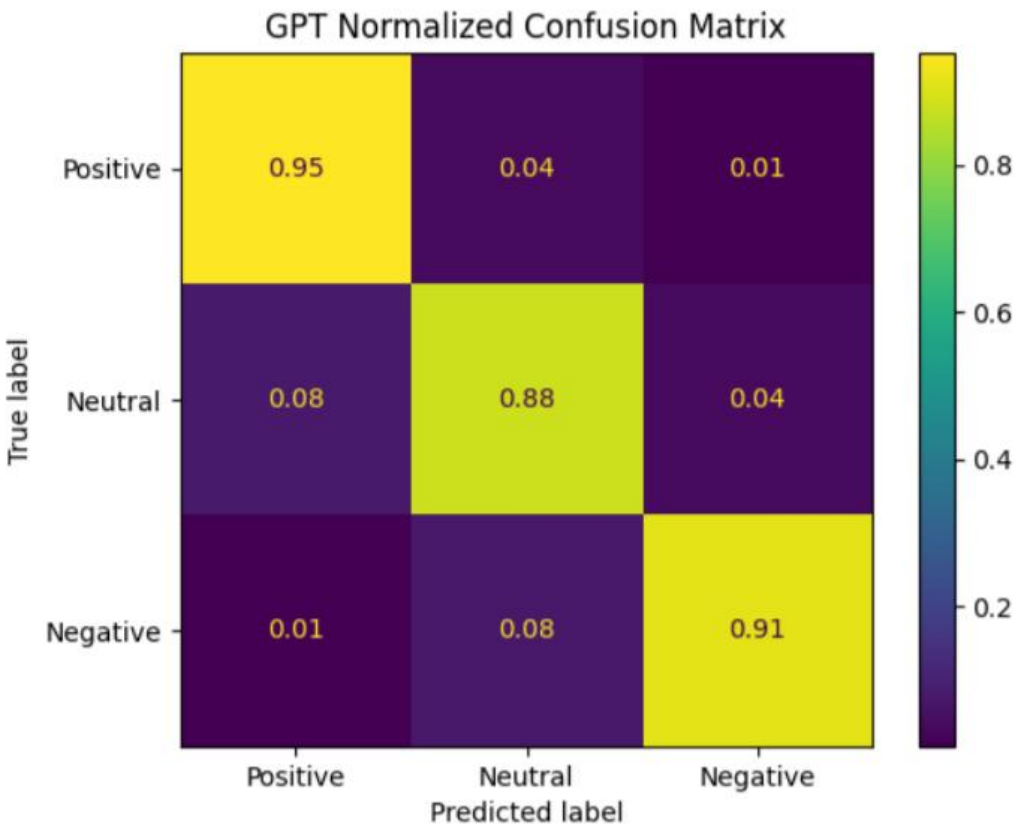
GPT Confusion Matrix



1. High Diagonal Concentration

The strong diagonal concentration in the confusion matrix indicates that GPT-4o-mini maintains high decision consistency across most samples, with over 90% of Positive and Negative cases remaining stable after paraphrasing.

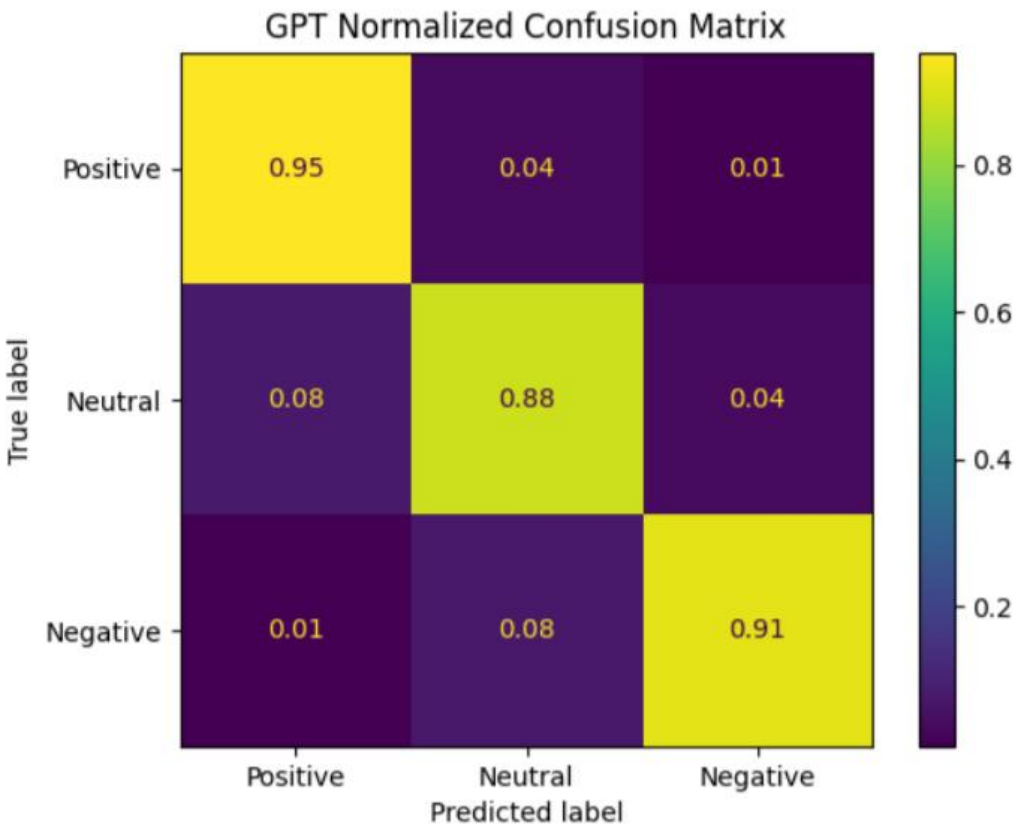
GPT Confusion Matrix



2. Neutral Class Vulnerability

Consistent with previous findings, the "Neutral" category exhibits the lowest stability (88% agreement). The misclassifications are predominantly directed toward adjacent categories (Positive or Negative), confirming that neutral sentiment acts as a volatile decision boundary.

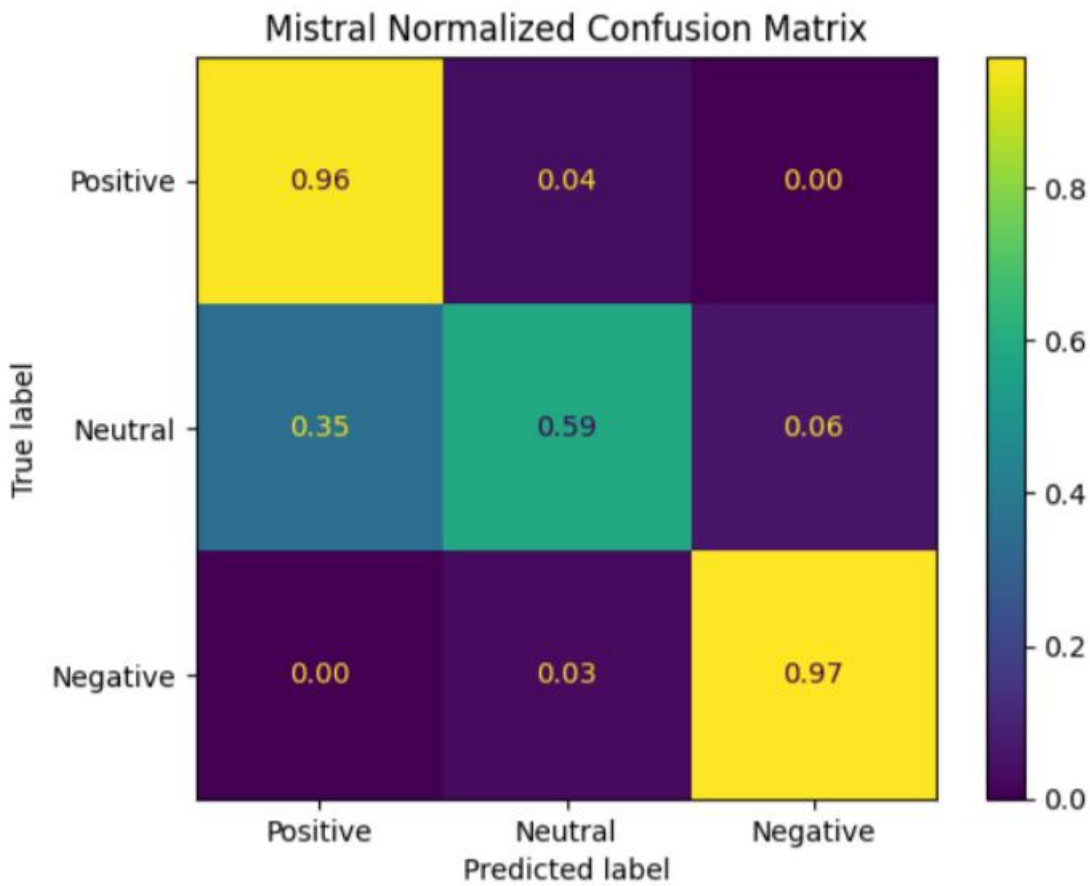
GPT Confusion Matrix



3. Rare Polar Reversals

Direct flips between "Positive" and "Negative" are extremely rare (approx. 1%), suggesting that while paraphrasing can shift the perceived intensity of an emotion, it seldom reverses the fundamental valence of the sentiment.

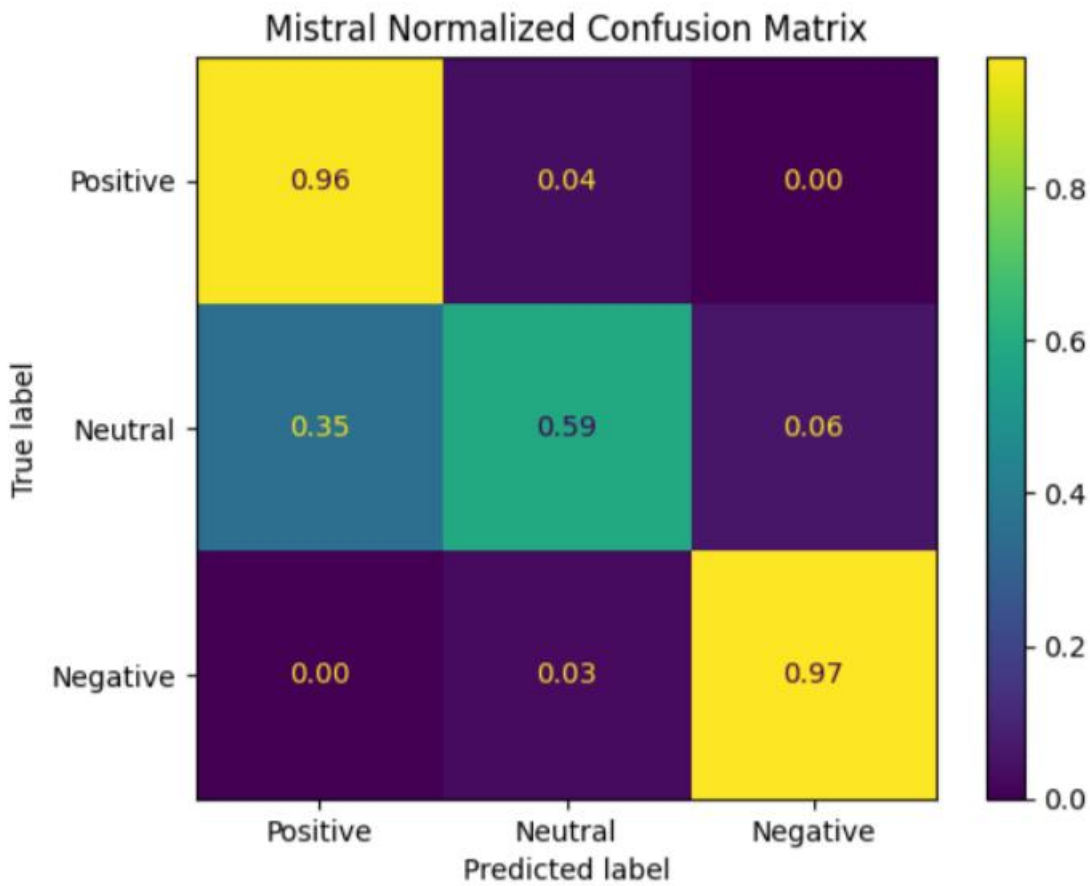
Mistral Confusion Matrix



1. Polarity Stability

Mistral demonstrates superior stability in "Positive" (96%) and "Negative" (97%) categories, even slightly outperforming GPT-4o-mini. This indicates a high level of robustness when dealing with explicit emotional valence.

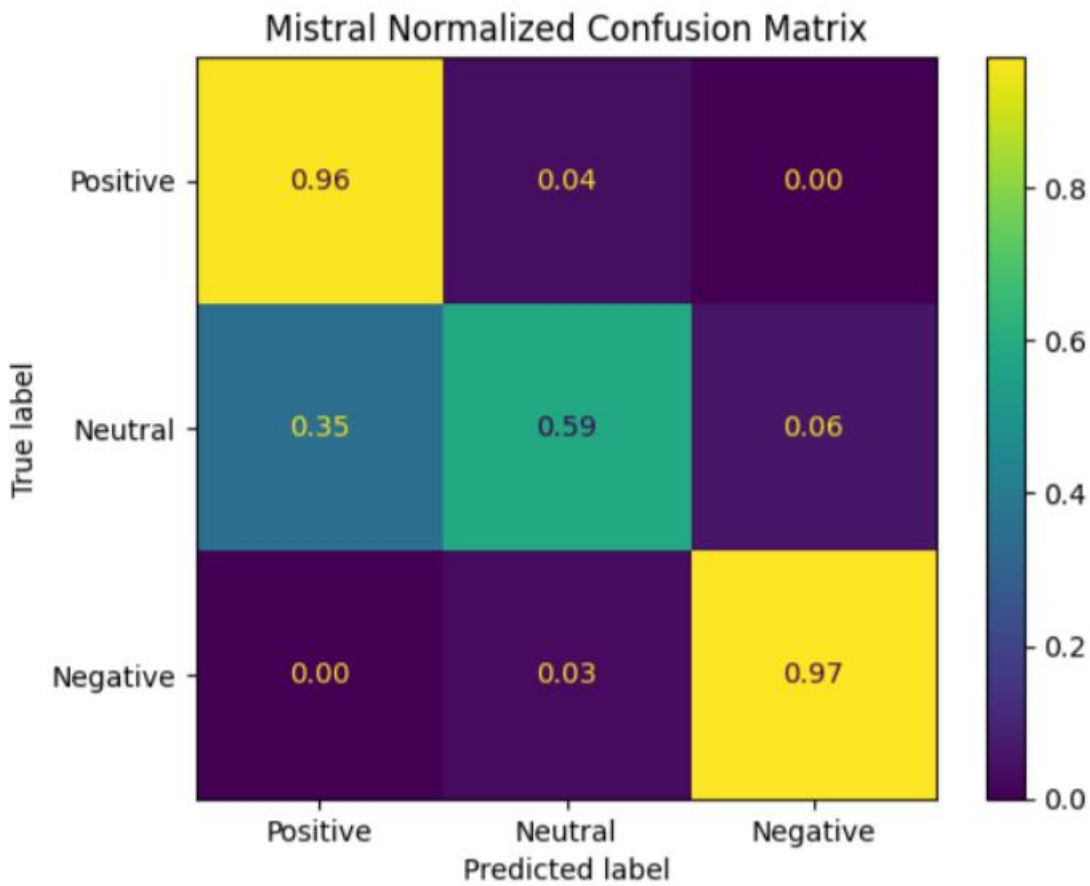
Mistral Confusion Matrix



2. Neutral-to-Positive Bias

The "Neutral" category is a major failure mode for Mistral, with a stability rate of only 59%. A significant portion of neutral samples (35%) are flipped to "Positive" after paraphrasing, suggesting a strong optimistic bias in the model's decision boundary.

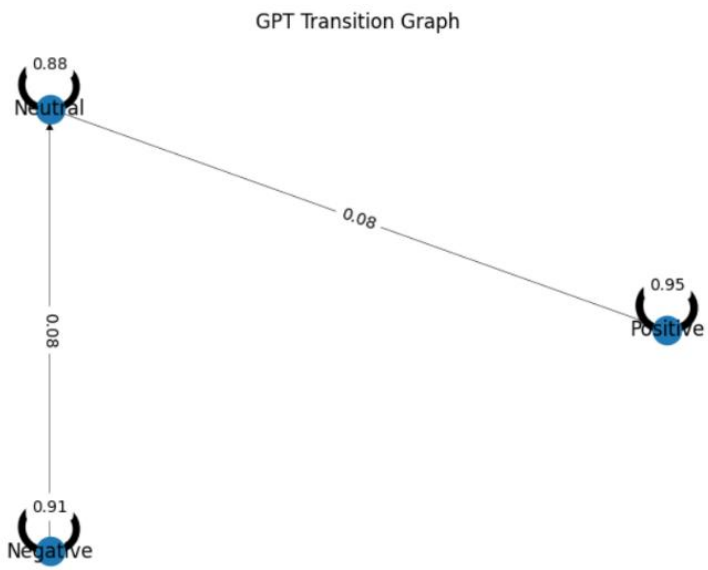
Mistral Confusion Matrix

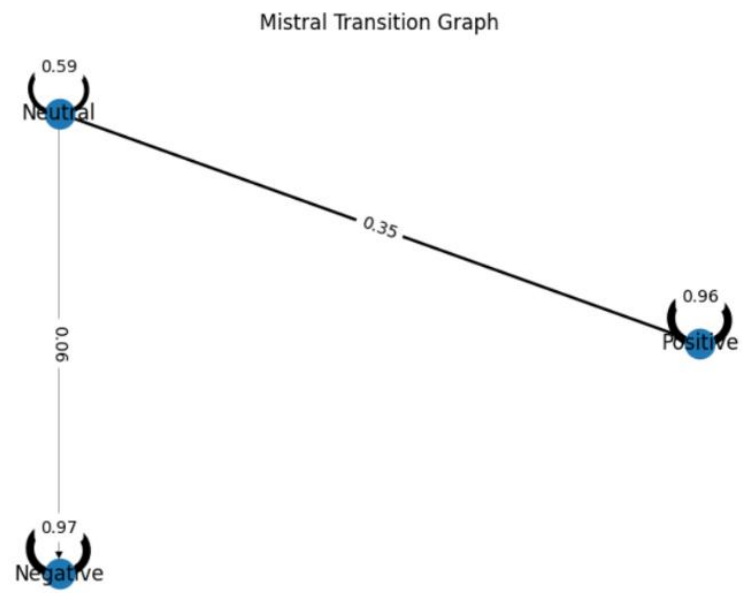


3. Robust but Brittle Boundaries

While Mistral effectively avoids cross-polar flips (0% between Positive and Negative), its high sensitivity to neutral phrasing confirms the professor's observation that smaller models struggle significantly with classification boundaries.

Transition Graph






```
plt.title("GPT  Transition  Graph")
plt.show()
```

gpt_para_pred	Negative	Neutral	Positive
gpt_orig_pred			
Negative	0.913978	0.075269	0.010753
Neutral	0.040000	0.880000	0.080000
Positive	0.012195	0.036585	0.951220

1. High Stability in Polarized Sentiments and Neutral Class as the Primary Volatility Source: The Neutral node acts as a major decision boundary, exhibiting significant "leakage" where sentences are easily flipped to adjacent categories.

2. Directional Bias in Smaller Models: Mistral shows an optimistic Bias, reclassifying 35% of neutral reviews as positive, whereas GPT-4o-mini maintains a more balanced but still sensitive transition pattern.

Case Study

example sentence 1:

The food is good. Unfortunately the service is very hit or miss. The main issue seems to be with the kitchen, the waiters and waitresses are often very apologetic for the long waits and it's pretty obvious that some of them avoid the tables after taking the initial order to avoid hearing complaints.

The food is enjoyable, but the service can be quite inconsistent. The primary problem appears to be in the kitchen, as the servers frequently apologize for the lengthy delays, and it' s clear that some of them steer clear of the tables after taking orders to dodge complaints.

Negative

↓

Neutral

The original sentence contains a strong negative cue, "unfortunately", which makes the overall sentiment negative. After paraphrasing, the phrase becomes "can be ...", which weakens the negativity. This shows that paraphrasing can affect sentiment intensity.

gpt_flip_cases_with_shift.csv
18.39 KB

Download

	paraphrase	gpt_orig_pred	gpt_para_pred	gpt_orig_conf	gpt_para_conf	shift
y looking at the review for LJS? Its fried stuff, with extra fried stuff in a You will regret eating it 30 minutes after you finish, but it is delicious.	Are you seriously checking out the LJS review? It's just fried food, piled with more fried food in a cardboard container. You'll feel regret half an hour after eating it, but it tastes amazing.	Positive	Negative	0.8	0.8	Positive → Negative

Case Study

example sentence 2:

A fun place and lots to do but a bit pricy. $20.00 will buy you a hundred points which will last you a little over an hour. I recommend the bowling game ,ball racing car game and connect 4 also there are several games that can be played in groups of 4

It's an enjoyable spot with plenty of activities, though it's somewhat expensive. Spending $20.00 gets you a hundred points, which will last just over an hour. I suggest trying the bowling game, the ball racing car game, and Connect 4, plus there are several games suitable for groups of four.

Neutral

↓

Positive

the original sentence uses "a bit pricy", which is relatively mild negative, weakens the positiveness.

In the paraphrase, it becomes "somewhat", which contains less negativeness.

This increase in sentiment intensity pushes the prediction from Neutral to Positive.

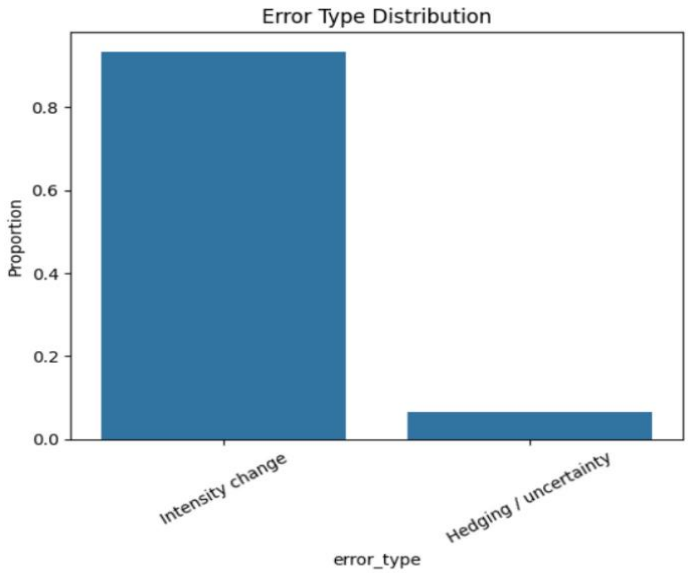
This indicates that small wording changes can move inputs across decision boundaries.

gpt_flip_cases_with_shift.csv
18.39 KB

Download

	paraphrase	gpt_orig_pred	gpt_para_pred	gpt_orig_conf	gpt_para_conf	shift
y looking at the review for LJS? Its fried stuff, with extra fried stuff in a You will regret eating it 30 minutes after you finish, but it is delicious.	Are you seriously checking out the LJS review? It's just fried food, piled with more fried food in a cardboard container. You'll feel regret half an hour after eating it, but it tastes amazing.	Positive	Negative	0.8	0.8	Positive → Negative

Error taxonomy



Most label flips are driven by changes in sentiment intensity, while only a small portion is related to hedging or uncertainty. This suggests that paraphrasing mainly affects how strongly sentiment is expressed, rather than changing its polarity or structure.

```
            model="gpt-4o-mini",
            temperature=0,
            messages=[
                {
                    "role": "user",
                    "content": f"""
You are analyzing sentiment classification errors.

Original sentence:
"{text}"

Paraphrased sentence:
"{para}"

The sentiment label changed after paraphrasing.

Classify the main reason into ONE of the following categories:

1. Hedging / uncertainty
2. Intensity change
3. Negation change
4. Ambiguity / borderline
5. Structural rephrasing
6. Other

Answer with only the category name.
"""
                }
            ]
        )
        return response.choices[0].message.content.strip()
```

Conclusion

First, label flips do exist, but they are not random. Both GPT and Mistral models show a consistent flip rate of about 7%, which means most of the time they are robust, but they aren't perfect.

Second, the instability is a boundary issue. Neutral or low-confidence samples are the 'danger zone' for flips. This confirms that the model’s decision boundaries are brittle when the sentiment is not strongly positive or negative.

Finally, sentiment intensity is the main driver. Subtle shifts in how a word is perceived can push the model over the edge.

Moving forward, developers could focus on 'Reliability Guidelines' for LLM deployment to ensure that a user’s choice of words doesn't accidentally change the model's critical decisions.

Thank you!

ST5230 Applied Natural Language Processing
Group21 Lin Ting

NUS
National University
of Singapore